BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 128 · Lesson 128 · Run-Length Encoding: Encoding and Decoding

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. When must lossless compression be used?

Q2. In a queue, what does FIFO stand for?

Q3. What is a function parameter?

Q4. In a BST, which side holds a smaller value?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

RLE — Encodes consecutive equal symbols as a symbol and count.

run — A consecutive sequence of identical symbols.

overhead — Extra data required by an encoding scheme.

lossless — Decoding reproduces the original exactly.

New learning

Identify runs — Group identical adjacent symbols, not identical symbols separated by others.

Encode — Record every run as count and symbol consistently.

Check cost — Count storage using an explicit format, including counts and delimiters.

Worked example

Encode AAAAABBCCC using (count,symbol) pairs; verify the result.

1. AAAAA | BB | CCC — Predict the first step from the prompt.

2. (5,A)(2,B)(3,C) — Explain the intermediate result or decision.

3. Decode original — AAAAA =5A, BB=2B, CCC=3C. Encoded result (5,A)(2,B)(3,C); decoded data restores AAAAABBCCC exactly.

We Do · Practise with guidance

Encode ABABAB as count-symbol pairs, and explain whether RLE helps.

A B A B A B

Predict the first step from the prompt.

Six single runs

Explain the intermediate result or decision.

Encoded size increases

Six runs of length 1 give (1,A)(1,B) repeated three times. A simple pair encoding uses 12 units for a 6-character input; RLE expands it, ignoring headers.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Apply RLE to AAAABBBCCAAAA using count-symbol pairs. [6]

Question 2

Decode (3,X)(1,Y)(2,X) and explain why runs must retain order. [5]

Question 3

A bitmap row has 40 alternating pixels. Explain why RLE might increase file size and identify a better-suited pattern. [8]

Exit Ticket · Answer before checking

RLE always reduces the size of any input.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. When exact reconstruction is necessary, such as executable code.

Q2. First in, first out.

Q3. An input used by a subroutine.

Q4. The left subtree.

You Do pages · Independent questions

Q1. (4,A)(3,B)(2,C)(4,A); one mark per correct run/length and two for correct ordered output.

Q2. XXXYXX [3]; order preserves original position, merging separated X runs changes sequence [2].

Q3. 40 runs of length one [2], count overhead expands stream [2], long blocks of identical pixels compress well [2], quantified or contextual comparison [2].

Exit Ticket · Model answer

False. Frequent symbol changes create many short runs and can increase size.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: